

Technical Evaluation: Windows Server 2025 on Supermicro X9QRi-F+

Overview of Windows Server 2025 Features

Windows Server 2025 is the latest Long-Term Servicing Channel (LTSC) release (generally available November 2024) and builds upon Windows Server 2022 with significant enhancements in virtualization, storage, security, and hybrid management ¹. Microsoft highlights “*major advancements across the board for Hyper-V, GPU integration, Storage Spaces Direct (software-defined storage), software-defined networking, and clustering,*” making Windows Server 2025 a strong platform for modern workloads requiring high performance and scalability ¹. Below is an overview of core features and improvements, especially in the areas of virtualization, storage, security, and remote management, that are relevant to deploying Windows Server 2025 on a Supermicro X9QRi-F+ based system.

Virtualization and Hyper-V Enhancements

- **Hyper-V Scalability:** Windows Server 2025 dramatically increases virtualization scale. A single host now supports up to 2,048 logical processors and 4 PB of physical memory (up from 512 LP / 48 TB in Server 2022) ². Individual Generation 2 VMs can be configured with up to 240 TB of RAM and 2,048 virtual CPUs ³. This far exceeds the hardware limits of the X9QRi-F+ (which maxes out at 4 CPUs and ~1 TB RAM) and ensures the OS can fully utilize all resources the board provides.
- **Enhanced Hyper-V Features:** Several new Hyper-V capabilities improve performance and flexibility. **Dynamic Processor Compatibility** mode has been improved for clusters – it now dynamically uses the maximal CPU feature set common to all hosts, yielding better VM performance when live-migrating across different CPU generations ⁴ ⁵. **Accelerated Networking (AccelNet)** simplifies SR-IOV networking for VMs, automatically configuring host networking to use the high-performance SR-IOV datapath and thereby reducing VM network latency, jitter, and host CPU overhead ⁶ ⁷ (note: SR-IOV requires a capable NIC; see hardware notes below). By default, new VMs created via Hyper-V Manager are now Generation 2 for better security and performance ⁸.
- **GPU Partitioning and AI Workloads:** Windows Server 2025 introduces improved GPU virtualization. Through **GPU Partitioning (GPU-P)**, a physical GPU can be partitioned and shared across multiple VMs, each VM getting a dedicated fraction of the GPU’s resources ⁹. New high-availability support allows a VM using GPU-P to fail over to another node, and **live migration** is supported for GPU-partitioned VMs, enabling movement of GPU-accelerated workloads without downtime ¹⁰. These features are aimed at AI/ML and graphics workloads. (On the X9QRi-F+, GPU-P would require installing a modern GPU since the board’s onboard video is a basic 2D controller.)
- **Hyper-V Security: Hypervisor-Enforced Paging Translation (HVPT)** is a new security feature that protects critical kernel page tables from modification by malicious code, helping prevent certain buffer overflow attacks ¹¹. It is enabled by default on supported hardware. Additionally,

Virtualization-Based Security (VBS) enclaves are introduced, allowing applications to run sensitive code in isolated enclaves protected by the hypervisor ¹² . These features strengthen isolation but may require modern CPU support (see compatibility section for how this relates to older Xeon E5 processors).

- **Clustering and Availability:** Windows Server 2025 improves clustering for Hyper-V. Notably, it supports **Hyper-V Workgroup Clusters** – failover clusters where hosts are not domain-joined, simplifying deployments in edge or small environments ¹³ . Integration with Azure Stack HCI and Azure Arc also facilitates hybrid clustering. Failover Cluster networking is streamlined via **Network ATC (Autonomic Transfer Control)**, an intent-based networking feature that auto-configures host NICs for cluster roles (management, storage, etc.) to ensure consistent, optimized network settings across nodes ¹⁴ ¹⁵ .

Storage and Data Management Improvements

- **Faster Storage I/O (NVMe Optimizations):** Windows Server 2025 is tuned for high-speed NVMe SSDs. In fact, Microsoft reports that on identical hardware, Server 2025 delivers **up to 60% higher storage IOPS throughput** compared to Windows Server 2022 ¹⁶ . This is achieved through kernel optimizations that reduce NVMe latency and CPU overhead ¹⁷ . On a Supermicro X9QRi-F+, which natively offers 6 SATA ports (2 × SATA3 6Gb/s + 4 × SATA2) ¹⁸ ¹⁹ , using NVMe would require a PCIe adapter. While the board itself doesn't have M.2 slots, it does support NVMe drives via PCIe – and Windows Server 2025 will leverage those drives more efficiently than previous OS versions.
- **Resilient File System (ReFS) Enhancements:** Server 2025 brings major updates to ReFS aimed at performance and space efficiency. **Block Cloning** is now supported on ReFS (notably used by the new Dev Drive feature), which dramatically speeds up file copy operations by copying data block references in metadata instead of reading/writing the actual blocks ²⁰ ²¹ . This yields faster large file copies and reduced I/O – beneficial for virtual disk cloning, backups, or VM provisioning. Additionally, ReFS gains **native data deduplication and compression** support, allowing storage of redundant data more efficiently on ReFS volumes ²² . This brings ReFS closer to feature-parity with NTFS for general file server use.
- **Storage Spaces Direct and Replica:** In clustered storage, **Storage Spaces Direct (S2D)** is improved with **thin-provisioned volumes**. Administrators can now create S2D volumes that only consume physical capacity as needed, and even convert existing fixed-provisioned volumes to thin, reclaiming unused space back to the pool ²³ . This helps avoid over-allocation of storage in cluster scenarios. **Storage Replica** (for disaster recovery) gets performance boosts: it now supports **compression** of replication traffic to reduce bandwidth use ²⁴ , and an **Enhanced Log** mechanism that bypasses some file system overhead, improving block replication throughput and latency ²⁵ .
- **Server Message Block (SMB) Updates:** The SMB file-sharing protocol in Windows Server 2025 includes support for the efficient **LZ4 compression** algorithm, in addition to existing algorithms (LZ77, Huffman, etc.) ²⁶ . Compressed SMB can improve file transfer speeds on slower networks by sending less data. More notably, Server 2025 includes **SMB over QUIC** (an SMB transport over UDP/443 with encryption) for secure, VPN-less access to file shares over the Internet ²⁷ . When enabled, SMB over QUIC allows mobile or remote clients to securely connect to an on-prem file server using TLS 1.3 encryption, with firewall-friendly UDP ports – essentially bringing Azure Files-like connectivity

to Windows Server. This can be useful if the X9QRi-F+ server is used as a file server that needs to be accessed remotely in a secure manner.

Security Features and Hardening

Windows Server 2025 is a “**security-first**” release with multiple layers of hardening, building on the Secured-core initiative (combining TPM, Secure Boot, VBS, etc.) and introducing new protections:

- **Credential Guard by Default:** Microsoft’s Credential Guard (which uses virtualization-based isolation to protect LSASS secrets) is now enabled by default on capable systems ²⁸ ²⁹. This means on hardware that meets requirements (typically UEFI, TPM 2.0, and supporting CPU virtualization features), Windows Server 2025 will automatically isolate credentials to mitigate theft of hashes/tickets by malware. On the X9QRi-F+, however, this may not activate since the board does not support TPM 2.0 (discussed later) and the older CPUs might lack certain virtualization extensions for virtualization-based security.
- **Boot and Firmware Protection:** Windows Server continues to support **Secure Boot** to ensure the OS boots only trusted, signed bootloaders. The X9QRi-F+ features a UEFI 2.3.1 firmware with Secure Boot capability ³⁰, so when running Server 2025 it is recommended to enable Secure Boot in the BIOS to guard against bootkits and unauthorized Option ROMs. Additionally, the server supports **UEFI firmware** (ACPI 4.0, SMBIOS 2.7), which aligns with modern Windows requirements ³⁰. (Legacy BIOS boot is also supported on this board, but using UEFI is preferable for Windows Server 2025 to leverage modern security features.)
- **Virtualization-Based Security (VBS) Enclaves:** A new feature called VBS enclaves allows applications to run sensitive code in secure enclaves isolated by the hypervisor ¹². In essence, even if an admin or kernel is compromised, the enclave’s data remains protected. This is similar in spirit to technologies like Intel SGX or AMD SEV, but implemented via Windows hypervisor. Use of VBS enclaves may require specific CPU support (e.g. Mode Based Execution Control (MBEC) or other second-level translation features). The Intel Xeon E5-4600 v2 CPUs (Ivy Bridge architecture) do support VT-x and EPT, but lack newer features like MBEC and have no SGX; thus, while core virtualization will work, certain VBS functionalities might not be available or might run with reduced security on this platform.
- **Active Directory Security Improvements:** If the X9QRi-F+ server will function as a domain controller, note that Active Directory in Windows Server 2025 gets some notable updates. There is an **optional 32K directory database page size** (vs. 8K legacy) which greatly raises various AD object limits (for example, allowing ~3,200 values in multivalued attributes vs ~1,200 before) ³¹ ³². This can improve AD scalability in large enterprises. There are also improvements to AD object recovery and **confidential attribute** handling, and the introduction of **Delegated Managed Service Accounts (dMSA)** which eliminate manual password management for service accounts by having AD manage them automatically ³³. These are advanced AD features that enhance security and manageability in enterprise scenarios.
- **Other Security Features:** Windows Server 2025 integrates **Windows Defender Advanced Threat Protection** and other ATP capabilities for malware detection and endpoint security (cloud-delivered) ³⁴. It also includes **LAPS (Local Administrator Password Solution)** built-in, which automatically

rotates and backs up the local admin account password to Active Directory for each server ³⁵. SMB security is beefed up with new defaults that prevent NTLM relay attacks and brute-force password attempts on SMB shares ²⁷. Many of these features work out-of-the-box to harden the server against common attack vectors.

Remote Management and Hybrid Integration

Managing Windows Server 2025 is made easier both locally and especially in hybrid cloud scenarios, which is valuable for a system like the X9QRi-F+ that might be deployed on-premises but managed remotely:

- **Windows Admin Center & Azure Arc:** Windows Server 2025 has deep integration with Azure Arc for hybrid management. A new “*Azure Arc Setup*” wizard comes pre-installed (as a Feature on Demand) to onboard the server to Azure Arc easily ³⁶. Once connected, you can use **Windows Admin Center (WAC) in the Azure portal** to manage the server’s roles, features, and settings through a web GUI ³⁷. This essentially brings the familiar WAC interface into Azure, providing a single pane for managing on-prem and cloud servers together ³⁷. For an X9QRi-F+ host, this means you can perform remote management (e.g., view performance, configure features, manage VMs, etc.) from anywhere via Azure’s secure portal, even if the server itself is behind a firewall, as long as it’s Arc-connected.
- **Azure Hybrid Services:** With Arc integration, Server 2025 also offers **pay-as-you-go licensing** and other cloud-based benefits ³⁸. While licensing is beyond our scope, the key point is that connectivity to Azure enables features like Azure Site Recovery (for VM replication to cloud) ³⁹, cloud-based **monitoring and best-practice assessments** (Arc can scan the server and suggest configuration improvements) ⁴⁰, and even **just-in-time remote support** where Microsoft support personnel can, with permission, get time-limited access to assist with issues ⁴¹. These capabilities can extend the lifecycle and support of older hardware like the X9QRi-F+ by integrating it with modern cloud management frameworks.
- **PowerShell and Automation:** Windows Server 2025 includes **PowerShell 7** (the latest cross-platform, modern PowerShell) for improved scripting and automation ⁴². This allows admins to use advanced modules and scripts to manage the server both locally and remotely. Remote administration via PowerShell Remoting (over WinRM/SSH) is fully supported, and the inclusion of OpenSSH server by default (a feature carried over from Server 2019/2022) means you can securely SSH into the server for management tasks ⁴³.
- **Built-in Remote Management Tools:** The OS still supports traditional tools like Remote Desktop, Server Manager, and the new Windows Admin Center (which can also be installed locally if not using Azure Arc). In Windows Server 2025’s desktop experience (GUI mode), the environment is similar to Windows 11’s interface ⁴⁴, and tools like the modern **Windows Terminal** and **WinGet** package manager are available by default for administrator convenience ⁴⁵ ⁴⁶. This makes local and remote management more streamlined.
- **Out-of-band Management (Hardware):** Separately from the OS, it is worth noting the Supermicro X9QRi-F+ includes a dedicated Baseboard Management Controller (BMC) with **IPMI 2.0 + KVM-over-LAN** support ⁴⁷. This allows hardware-level remote management – you can power cycle the machine, access BIOS, and use a remote console independent of the OS. When planning remote

management, leveraging the BMC (via Supermicro's IPMIView or web interface) in conjunction with the OS tools provides comprehensive control. For example, if Windows becomes unresponsive, the IPMI can be used to reboot or troubleshoot at the firmware level.

Performance Characteristics of Windows Server 2025

Windows Server 2025 is designed to deliver robust performance on modern hardware, and it introduces several under-the-hood improvements in memory management, CPU scheduling, I/O throughput, and networking. When running on the Supermicro X9QRi-F+ platform (which is an older but high-end 4-socket server board), these OS enhancements will generally improve efficiency, though the extent will be bounded by the hardware's capabilities (DDR3 memory, older Xeon cores, limited SATA bandwidth, etc.). Below is an analysis of the OS performance characteristics in key areas:

Memory Management and Scalability

The memory subsystem in Windows Server 2025 can scale to **extremely large RAM sizes (up to 4 petabytes)**, ensuring that even memory-intensive workloads are handled with ease ². The OS includes refined algorithms for memory allocation, NUMA awareness, and caching. In a multi-socket system like the X9QRi-F+ (which has a NUMA node per CPU socket), Windows will automatically distribute and balance memory allocations per NUMA node to optimize local memory access. The OS's support for **disaggregated memory and huge memory pages** has been improved, which can help in HPC or virtualization scenarios by reducing overhead. For example, Hyper-V on Server 2025 can employ large memory pages more aggressively, and features like **Dynamic Memory** (if using Hyper-V) are expected to work more efficiently for guest VMs, adjusting allocations with less fragmentation.

On the Supermicro X9QRi-F+, the maximum memory is 1 TB (with 32 DDR3 RDIMMs) ⁴⁸ ⁴⁹. While DDR3-1600/1866 memory has lower bandwidth than newer DDR4/DDR5, Windows Server 2025's memory manager will utilize all available memory for caching disk operations and handling large in-memory workloads as needed. The OS's ability to handle memory pressure is enhanced by tools like **DTrace**, which is now built-in and can help profile memory and CPU usage in real time ⁵⁰. Administrators can use such tools to fine-tune performance (for example, observing if the system is swapping or experiencing NUMA imbalances). In summary, Windows Server 2025 will fully utilize the X9QRi-F+'s memory capacity, and its improvements in scalability mean the OS itself won't be a bottleneck even as the hardware approaches its limits.

CPU Utilization and Scheduling

Windows Server 2025 introduces an optimized CPU scheduler and smarter resource allocation, which is particularly beneficial for systems with many cores or multiple CPUs. The OS can handle up to 2048 logical processors, as noted, and while the X9QRi-F+ (with 4 × Xeon E5-4600 v2 CPUs) will present at most 48 physical cores / 96 logical threads, the scheduler improvements still help in this context. For example, the OS uses **CPU groups** to manage large core counts; in our case it won't need to split into groups, but it will leverage all cores for parallel tasks.

The scheduler also includes security-aware features (e.g., retpoline and hardware mitigations for Spectre/Meltdown are built-in), which do incur some overhead on older CPUs. The Xeon E5-v2 series does support speculative execution mitigations via microcode/OS, and Server 2025 will enable those by default to ensure

security. This can slightly reduce raw CPU throughput compared to an unpatched older OS, but it is a necessary trade-off.

That said, Windows Server 2025 has **enhanced CPU scheduling and resource allocation** algorithms that improve overall throughput for applications and services, making better use of each core ⁵¹ ⁴² . One specific feature, as mentioned earlier, is the **dynamic compatibility mode for processors** in Hyper-V clusters, which ensures VMs can use newer CPU features when available, without manual intervention ⁵² ⁵³ . On a single standalone server, this isn't directly applicable, but it indicates the OS is "aware" of CPU feature sets and can optimize usage accordingly.

In practice on the X9 platform, you can expect Windows Server 2025 to utilize all 96 logical processors efficiently under load. Workloads that spawn many threads (SQL Server, web servers, etc.) will be distributed across CPUs. The OS will handle the 4 NUMA nodes by trying to keep threads on their local memory node when possible (processor group assignments align with NUMA nodes when ≤ 64 cores each). If Hyper-Threading is enabled (it is on E5 by default), the scheduler is SMT-aware to avoid overloading sibling threads unnecessarily. Overall, the OS's modern scheduler should yield slight performance improvements over older Windows versions in terms of CPU throughput and responsiveness under heavy multitasking, even on this older hardware.

Storage I/O and Disk Performance

As noted, Windows Server 2025 brings significant storage performance gains, especially for fast disks. The **NVMe optimization** is a key improvement – the OS can achieve up to 60% more IOPS on the same NVMe SSD hardware compared to Server 2022 ¹⁶ . This is partly due to a more efficient Storport driver and mitigations for interrupt handling overhead. On the X9QRi-F+, if you use SATA SSDs or HDDs on the onboard Intel C602 SATA controller, the performance will mostly be limited by the 6Gb/s SATA3 interface. However, the OS will still take advantage of NCQ (native command queuing) and other storage stack improvements.

If high performance storage is needed, one could install a PCIe NVMe drive or an HBA/RAID card. Windows Server 2025 has in-box drivers for many common storage adapters and will benefit from the NVMe and RAID enhancements automatically. For example, a PCIe 3.0 NVMe on this board (though running at PCIe 3.0 x4 speeds) could see better I/O due to lower latency in the OS stack. Features like **multi-queue I/O** scheduling, which were introduced in Server 2016 and refined since, are at their best in Server 2025 – meaning the OS can efficiently handle many simultaneous disk operations by leveraging multiple processor cores for I/O.

The **ReFS block cloning** feature mentioned earlier also has performance implications: copying or cloning large files (such as VHD/VHDX virtual disks for VMs) is far faster and uses much less disk bandwidth now ²⁰ ²¹ . This can drastically shorten maintenance tasks like backing up VMs or duplicating datasets. Additionally, if deduplication is enabled (now possible on ReFS in 2025), read I/O for duplicate data is reduced and more of the working set might be served from cache, effectively boosting read performance for redundant data scenarios.

One thing to consider is that the X9QRi-F+ chipset (C602/X79 platform) is an older generation chipset – it supports SATA2/3 and some legacy PCIe 3.0 lanes. The Windows 2025 storage stack is fully backward-compatible with such hardware, but some new storage features (like Storage Spaces Direct with NVMe caching, or SMB Direct RDMA over RoCE) would require hardware this board doesn't have (NVMe drives,

RDMA NIC). In essence, basic disk I/O (local or SAN) will be as good as the hardware allows, and the OS improvements will ensure minimal overhead. Any existing performance bottlenecks will likely be at the hardware layer (e.g., the limits of DDR3 memory throughput feeding the I/O or the SATA 6G bus) rather than in the OS.

Network Throughput and Stack Efficiency

Windows Server 2025 includes a range of **network performance enhancements**, which can especially shine with modern high-speed network adapters. The OS has improved the TCP/IP stack – for instance, it has better default TCP congestion control and recovery (building on innovations like TCP HyStart++ and RACK loss recovery that were introduced in Windows 11/Server 2022). It also adds support for ultra-high-speed networks such as 25 GbE and 50 GbE, with tuned defaults to reduce CPU overhead at those speeds ⁵¹ ⁵⁴. Networking features like **Receive Segment Coalescing (RSC)** and **Virtual Receive Side Scaling (vRSS)** for VM workloads are all present and optimized.

On the Supermicro X9QRi-F+, the onboard NICs are dual **Intel i350 Gigabit Ethernet** ports ⁴⁷. These are capable and reliable NICs, but limited to 1 Gbps throughput per port. They do support some offloads (checksum offload, LSO, VMDq for virtualization), but they do **not support SR-IOV** or RDMA. This means advanced features like **AccelNet** (which automates SR-IOV for VMs) won't apply on the stock NICs – you would need to add, say, a 10GbE or 25GbE NIC that supports SR-IOV if you wanted to utilize that feature on this server. If a modern NIC is added via PCIe (the board has several PCIe 3.0 x8/x16 slots), Windows Server 2025 will take advantage of it fully: for example, a 10GbE Intel X540 or X550 NIC could be used, and though the CPU is older, the OS would handle the higher interrupt rate by possibly utilizing features like **Receive Side Scaling (RSS)** across multiple cores to distribute network load.

Even with the 1GbE interfaces, the OS improvements are beneficial. The networking stack in Server 2025 has been hardened and made more efficient. For instance, default network security policies (if managed via WAC) block unwanted inbound traffic by default ⁵⁵ ⁵⁶, which can slightly reduce exposure and load. The inclusion of SMB compression (LZ4) means that copying large files over the network (if both ends are 2025) could be faster on bandwidth-constrained links – although on a 1 Gbps LAN, the benefit may be limited by CPU compression speed (the Xeon E5 v2 does have AES-NI and is reasonably powerful per core, so LZ4 at 1 Gbps should be fine).

Another improvement is **Network ATC** for cluster networks (making it easier to configure QoS and VLANs for different traffic types) ¹⁴. In a single-server scenario this is less relevant, but if the X9QRi-F+ is part of a failover cluster, Network ATC can ensure that say storage replication traffic vs VM traffic are separated and optimized with minimal admin effort.

Overall, Windows Server 2025's network stack is highly optimized: lower latency, better multi-core scalability for packet processing, and enhanced security. On an older platform, you might not be able to exploit the highest-end capabilities (like 50GbE NIC offloads or RDMA), but you will still get a rock-solid networking foundation that is more efficient than previous Windows Server versions. Expect that the dual Intel i350 NICs will be able to run at near line-rate 1 Gbps with low CPU usage (~5-10% of one core typically, thanks to offloads), and if you team/bond them or use both, the OS can utilize both ports (e.g., with SMB Multichannel for 2 Gbps aggregate SMB throughput). If needed, upgrading to a faster NIC is straightforward and supported by the OS with improved performance at those higher throughputs ⁵¹ ⁵⁴.

Compatibility and Installation on Supermicro X9QRi-F+

The Supermicro X9QRi-F+ is a quad-socket server motherboard (approx. circa 2013) that supports Intel Xeon E5-4600 and E5-4600 v2 series processors (up to 12 cores per CPU) and uses the Intel C602 chipset ⁵⁷ ⁵⁸ . Installing Windows Server 2025 on this board is technically feasible and generally runs well, but there are a few compatibility and setup considerations to ensure a smooth experience. This section covers chipset/CPU support, BIOS requirements, driver availability, and other installation notes specific to this platform.

Chipset and CPU Support

- **CPU Architecture:** The X9QRi-F+ supports the Intel Xeon E5-4600 v2 family (“Ivy Bridge-EP”, 22nm) and v1 (“Sandy Bridge-EP”) in LGA2011 sockets ⁵⁷ . These CPUs are 64-bit, support the required instruction extensions (Intel 64, NX bit, VT-x, VT-d, etc.), and in principle can run Windows Server 2025. However, it’s important to note that **these older Xeon E5 processors are not on Microsoft’s official supported CPU list for Windows Server 2025**. Microsoft’s certified processor list for Server 2025 starts at much newer generations (e.g. Intel Scalable Xeons Gen2+ and AMD EPYC 7xx2+); pre-2015 CPUs like the Xeon E5-v2 are not included ⁵⁹ ⁶⁰ . This does **not** mean it won’t work – it simply means Microsoft won’t test or guarantee Windows 2025 on such old silicon for new deployments. In practice, many admins have reported success running Server 2022 and even 2025 on Xeon E5-v2 systems. The Reddit community, for example, found that a dual Xeon E5-2690 v2 system could run Windows Server 2025 (used as a desktop OS to bypass Windows 11’s restrictions) without major issues ⁶¹ . So compatibility is generally fine, but be aware you are in “unsupported” territory.
- **Performance and Features:** Because the Xeon E5-4600 v2 lacks some modern CPU features, a few Windows 2025 capabilities will be limited. For instance, as discussed, **Credential Guard and VBS** might not enable due to missing second-level address translation features like MBEC. The processors do have first-generation Intel VT-x with EPT and VT-d for virtualization, which *are* sufficient for running Hyper-V and basic VBS isolation. They also have AES-NI for encryption, so features like BitLocker and SMB encryption will use hardware acceleration. Overall, the OS will utilize the CPUs fully, but just note that the raw performance is that of 2013-era 2.0–2.9GHz cores with no turbo beyond 3.x GHz and no AVX2 instructions. Modern workloads (especially ones using heavy AVX2/AVX-512 on newer CPUs) will run, but slower. That said, standard server tasks (AD DS, DNS, file services, moderate virtualization) should run comfortably within the CPU capacity.
- **Chipset (Intel C602):** The C602 server chipset on the X9 provides the SATA ports, USB, PCIe lanes, etc. Windows Server 2025 includes in-box support for the Intel C600 series chipset. Basic devices like the PCIe root ports, USB 2.0 controllers, and SATA AHCI controller will be recognized with Microsoft’s standard drivers. If the SATA controller is set to **AHCI mode**, the install should see any connected drives natively. If it’s in **Intel RSTe RAID mode** and you have a RAID volume defined, you might need to load the Intel RSTe driver at install (Windows 2025 may not include the out-of-the-box driver for this older RAID controller). Intel last provided drivers for RSTe (Rapid Storage Tech Enterprise) for Windows Server 2016/2019, but those drivers often still work on 2022/2025. Alternatively, using AHCI mode is simplest unless you require a firmware RAID.
- **Memory and NUMA:** No compatibility issues here – the OS will see the memory installed. With four CPUs, the system presents a NUMA topology that Windows will recognize and optimize for. One

consideration is to ensure memory is populated per the board's guidelines (all four CPUs should have memory, ideally balanced across channels) to avoid any CPU running without local memory. The board can support up to 1 TB of RAM (with 32 GB RDIMMs) ⁴⁸, which is far below the OS's max, but still plenty. Windows Server 2025 requires a minimum of 2 GB (4 GB recommended for Desktop Experience) ⁶², so obviously that's not a problem here.

BIOS/UEFI and Firmware Considerations

- **BIOS Version:** It is **highly recommended to update the X9QRi-F+ to the latest BIOS firmware** before installing Windows Server 2025. Supermicro released BIOS updates for this board up through mid-2015 (to support the v2 processors and fix bugs). The latest revision (often version 3.x or similar) should include the latest CPU microcodes and stability fixes. Updated microcode is important for security (Spectre/Meltdown mitigations) and stability under heavy loads. Administrators have noted that BIOS updates are a first step to resolve any weird issues on Supermicro platforms ⁶³. For example, if you encounter instability or crashes, ensure the BIOS is current and loaded with optimal defaults.
- **UEFI vs Legacy Boot:** The X9QRi-F+ features a UEFI BIOS (conforming to UEFI 2.3.1 and supporting ACPI 4.0) ³⁰. Windows Server 2025 can install in either UEFI or Legacy (CSM) mode. **UEFI mode is strongly recommended** for a few reasons: it allows use of GPT partitioned boot drives (needed if your boot volume is over 2 TB), and it enables Secure Boot. If you install in legacy BIOS mode, you won't be able to enable Secure Boot and might run into the MBR 2 TB boot disk limit. In the BIOS setup, ensure the boot mode is set to UEFI. You can also enable Secure Boot (load the default keys provided by Supermicro) to take advantage of OS boot chain verification ⁶⁴. Secure Boot on this board should work since it's listed as supported, guarding against boot-time malware by verifying signatures of the boot loader ⁶⁴.
- **TPM (Trusted Platform Module):** A notable compatibility point is the lack of TPM 2.0 support on this motherboard. The X9 generation boards pre-date TPM 2.0; Supermicro did offer optional TPM modules (via header) for some X9 boards, but they were TPM 1.2 modules. In fact, users have reported that *"TPM 2.0 module for this motherboard doesn't exist... [you] would need a Supermicro X10 or higher"* to have TPM 2.0 ⁶⁵. This means if you require TPM 2.0 for features like BitLocker with TPM unlocking, VBS with Credential Guard, or generally to satisfy modern OS recommendations, you're out of luck. Windows Server 2025 does **not** actually require a TPM to install or run (unlike Windows 11 which mandates TPM 2.0 for official support), so the OS will still function on the X9QRi-F+ without one. But certain security features will fall back or be disabled. For example, without TPM 2.0, Credential Guard cannot store its secrets in hardware, and the system cannot be a "secured-core server" by Microsoft's definition. If security is a priority, you might consider using an older TPM 1.2 module if available (it could at least be used for BitLocker encryption, albeit without the enhanced algorithms of TPM 2.0).
- **NVMe Boot:** As this board was released before NVMe drives were common, its BIOS does not natively support booting from NVMe SSDs (no NVMe DXE driver present). If you plan to use an NVMe drive via a PCIe adapter as your OS disk, you have a couple of options: 1) Use the **Clover or DUET** UEFI NVMe driver hack, or a modded BIOS that adds NVMe support (enthusiast forums like Win-Raid have mods for X9 boards) ⁶⁶, or 2) Simpler, use a small SATA SSD for the OS, and use NVMe for secondary storage. The second approach is often easier – install Windows on a SATA drive connected

to one of the onboard ports (or a RAID volume that the BIOS can boot from), then once Windows is running it will see the NVMe drive and you can use it for data, VMs, etc. Windows Server 2025 will definitely detect and use NVMe drives (the OS has an in-box NVMe driver). It's purely a BIOS boot limitation to be mindful of.

- **BMC/IPMI Firmware:** It's also wise to update the IPMI firmware (BMC firmware) to the latest version from Supermicro's site. The X9QRi-F+ uses a Nuvoton WPCM450 BMC, which has had some security updates over the years. Updating it can improve remote management reliability and patch known vulnerabilities. The IPMI operates independently of Windows, but a stable BMC ensures that things like sensor readings and fan control work properly alongside the OS. There have been cases where outdated BMC firmware caused sensor interrupts or IPMI driver conflicts in OSES, so keeping it updated is part of ensuring overall system stability.

Driver Availability for Onboard Devices

One advantage of running a new OS like Windows Server 2025 on older, well-supported hardware is that most devices have mature drivers that often come bundled with the OS. Here's what to expect for the X9QRi-F+ onboard components:

- **Network Controllers:** The board features dual Intel Gigabit LAN controllers (Intel® i350-AM2 dual-port NIC) ⁴⁷. Windows Server 2025 should have native drivers for the i350 – Intel NIC support is built into Windows, especially for such ubiquitous chips. Upon installation, the two NIC ports will likely be recognized as “Intel(R) Ethernet Server Adapter I350” or similar and be usable immediately. For optimal performance, you can install the latest Intel PROSet driver package (if one exists for Windows 2022 it will likely work on 2025 as well). However, basic functionality (IPv4/IPv6, VLANs, teaming via LBFO or SET, etc.) will work out-of-box. The IPMI dedicated LAN port uses a Realtek PHY, but that is only for the BMC – it does not expose a NIC to the OS, so no driver needed for that.
- **Storage Controllers:** The six onboard SATA ports (2 × 6Gb/s + 4 × 3Gb/s) are provided by the Intel C602 chipset's AHCI controller ¹⁸ ¹⁹. Windows will use the standard AHCI driver unless you've configured a RAID. If using chipset RAID (Intel RSTe), you might need the Intel RSTe driver as mentioned. If you have additional storage like an LSI/Avago MegaRAID or HBA card in a slot (common in 24-bay server setups with this board), you'd need the driver for that card – but again, most LSI SAS2008/2308 or SAS3xxx controllers have in-box support or easily obtainable drivers for modern Windows. For example, LSI 9211-8i cards are supported by the Microsoft “StorPort” driver out-of-box. It's always good to check device manager after install for any “unknown device” – but typically on X9 boards, the only unknown device might be the IPMI interface (see next point) or SMBus, which is usually handled by installing the **Intel Chipset INF** package (which simply labels the chipset devices but doesn't actually add functionality).
- **IPMI Interface Driver:** The Supermicro BMC exposes a KCS interface for IPMI-over-PCI. If you want to use in-band IPMI management or monitoring from the OS (for example, using IPMIutil or ipmitool to read sensors, or if using something like the Supermicro Power Manager software), you may need to install the **Microsoft IPMI driver** (which is an optional feature in Windows – in Server 2025, you can likely add it via Server Manager -> Features -> “IPMI Interface”). This isn't strictly necessary for most users, since the BMC can be managed via network independently. But it's worth noting in case a monitoring application expects it.

- **Onboard Video:** The motherboard's video output is driven by the BMC's integrated graphics (Matrox G200 core). Windows will usually use a basic display driver (Microsoft Basic Display Adapter) for this, which is fine for server purposes. You won't get high resolution or 3D acceleration, but it's enough for installation and troubleshooting. If you plan to use the server with a GUI and higher resolution, you can install a basic Matrox driver or install an external GPU. However, given this is a server, most likely the onboard video will suffice (especially since IPMI allows headless use via remote console).
- **USB Ports:** The X9QRi-F+ has USB 2.0 ports (eight in total: 4 rear, 3 front header, 1 Type-A internal) ⁶⁷ ⁶⁸ . These are UHCI/EHCI controllers from the chipset, and Windows will automatically use them (the installer will run from a USB drive just fine). There are no USB 3.0 ports, so no need for USB 3 drivers.
- **Misc Devices:** Supermicro boards often have extra serial ports, fan controllers, etc., but none of those typically require special drivers. The hardware monitoring (voltage/temp/fan sensors) is handled by the BMC. If you install Supermicro's SuperDoctor utility, it can communicate with the BMC to report sensors – again that might use the IPMI driver or direct network queries.

In summary, driver support in Windows Server 2025 for the X9QRi-F+ is very good. The core hardware (CPU, chipset, NIC, storage, USB) will work out-of-box or with minor INF installs. This platform was common in the Windows Server 2012 era, so by now its support in modern OSes is mature. The primary caution is simply that since the hardware is older, you may not find *Windows Server 2025-specific* drivers from Supermicro or Intel if you run into an issue – you'd rely on the last available drivers (e.g. Windows 10/Server2016 era drivers) which generally do work.

Hyper-V and Virtualization on the X9QRi-F+

Deploying Hyper-V or other virtualization solutions on the Supermicro X9QRi-F+ (Xeon E5-v2 platform) under Windows Server 2025 is quite feasible and can be effective for many workloads, but it comes with some caveats due to the age of the hardware. Here are key points and “notes from the field” regarding virtualization on this setup:

- **Hyper-V Role Compatibility:** Windows Server 2025 Hyper-V requires a 64-bit processor with Second Level Address Translation (SLAT). The Xeon E5-4600 v2 **does** support SLAT (Intel EPT), and Intel VT-x virtualization extensions, so it meets the technical requirements to run Hyper-V and host VMs. You'll need to enable “Intel Virtualization Technology” (VT-x) and “VT-d” in the BIOS if they aren't already on (on Supermicro boards they are usually enabled by default, but double-check BIOS settings for “Virtualization” and “VT-d”). With those on, the Hyper-V role should install without issue and you can create VMs that utilize the full CPU and memory of the machine.
- **Performance for VMs:** Each Xeon E5-46xx v2 CPU has up to 12 cores (24 threads). In a fully loaded configuration (4 CPUs, 12C each, HT on) you have 48 physical cores / 96 logical processors available to Hyper-V. Windows Server 2025 Hyper-V can handle this easily (far below its 2048 LP limit ²). The scheduler will distribute VMs across the NUMA nodes. By default, Hyper-V will treat each socket as a NUMA node with its local memory, and it will try to keep a VM's memory allocation within one node for performance (as long as the VM's vCPU count doesn't exceed the cores of a single socket, it will stay “NUMA local”). You can manually configure NUMA spanning settings if needed, but the defaults are fine.

- **Older CPU Feature Set:** One thing to note is that if you plan to run modern guest OSes (like Windows 11/Server 2025 or certain Linux distros) in VMs, you might need to enable **Processor Compatibility Mode** if you ever move those VMs to a host with a newer CPU, but on a single host this isn't an issue. The VMs will see a virtual CPU based on the host's capabilities. The E5-v2 lacks features like AVX2, so any software inside VMs that attempts to use AVX2 will just not have it available. This typically isn't an issue unless running heavy scientific workloads inside VMs.
- **Hyper-V Enhanced Security Features:** As mentioned under security, new Hyper-V related security features like HVPT and VBS enclaves may not be fully supported on this hardware. For example, **HVCI (Hypervisor-protected Code Integrity)** for VMs or the host might not be enabled because it requires Intel mode-based execution control (MBEC) or an equivalent, which older CPUs don't have. If you enable features like **Shielded VMs** or **Virtual TPM** for VMs, they require a physical TPM in the host (or a host guardian service) – and since X9QRi-F+ has no TPM 2.0, using Shielded VMs with virtual TPM would not be straightforward (you'd have to use an external Key Protection Service rather than TPM binding). In short, core Hyper-V works fine, but "Secured-core" Hyper-V features will be limited.
- **Device Passthrough (VT-d):** The platform supports VT-d, and Windows Server Hyper-V can use it for **Discrete Device Assignment (DDA)**, i.e., PCI passthrough of devices to VMs. You could, for instance, pass a PCIe storage controller or GPU directly into a VM. The Intel C602 chipset and BIOS support VT-d (make sure "VT-d" is enabled in BIOS). However, do note that DDA (passthrough) in Hyper-V has specific requirements and not all devices/drivers are compatible. But generally, an Intel i350 NIC could be passed to a VM if desired (though with only two NICs, doing so might leave the host with no connectivity except IPMI, so be cautious). Likewise, a GPU (if you install one) could potentially be assigned to a VM for CUDA or rendering tasks – although NVIDIA consumer GPUs often have artificial blocks against virtualization on server OS, whereas NVIDIA Grid/Quadro or AMD GPUs might work.
- **SR-IOV and Networking for VMs:** The Intel i350 NICs on this board do **not** support SR-IOV, so you cannot use the "Single Root I/O Virtualization" feature of Hyper-V on the built-in NICs. Hyper-V will simply use traditional virtual switching (Hyper-V Virtual Switch) and VMs will share the 1Gb links via the vSwitch. This works fine, albeit with the overhead of virtualization. The **Accelerated Networking (AccelNet)** feature in Windows Server 2025, which streamlines SR-IOV configuration ⁶, is moot on this hardware unless a SR-IOV capable NIC is added. If high network throughput or low latency for VMs is a goal, installing a PCIe 10GbE or 40GbE NIC that supports SR-IOV (e.g., Intel X710 or Mellanox ConnectX-4) would allow you to take advantage of those Hyper-V capabilities on this OS. Otherwise, expect roughly up to ~3-4 Gbps total throughput through a Hyper-V virtual switch on a 1 Gbps NIC (since multiple VMs have to share 1 G link, the limit is 1 Gbps per NIC).
- **Storage for VMs:** If using this server as a Hyper-V host, consider that the storage system (especially if just using SATA drives) might be the performance bottleneck for VM I/O. With Windows Server 2025, using **VHDX** format is recommended (for features like block-level provisioning and online resizing). The ReFS improvements (block cloning) particularly help with creating or checkpointing VHDX files ²⁰ – for example, merging a checkpoint or copying a VHDX could be much faster than on older OS versions. If you have multiple drives, you can use **Storage Spaces** to create a simple or mirror pool for the VMs; Windows 2025 will support that just as 2019/2022 did. The X9 board itself has no built-in SAS expander or HBA, so in a large drive configuration an add-in card is used – ensure

any such card is compatible (most LSI RAID/HBAs are). The OS can also do **Storage Replica** in case you want to replicate VM storage to another server for disaster recovery; just remember the 1 Gbps NICs would limit replication speed unless upgraded.

- **Scaling and Overcommitment:** With up to 96 logical processors, you can run a fair number of VMs on this host. Windows Server 2025 Hyper-V supports huge VM sizes (as noted), but on this hardware you might run, say, 20–30 moderate VMs (depending on CPU/RAM needs) before hitting resource limits. Features like **Memory overcommit (Dynamic Memory)** can be used to maximize utilization, and Windows 2025's enhancements in memory management will help avoid issues when overcommitting. The **workgroup cluster** feature could even let you set up a two-node Hyper-V cluster using another similar server without needing a domain ¹³, which is an interesting option for high availability in a lab or small biz setting – though the X9 boards might be too large and power-hungry for a small cluster unless you already have them on hand.

In summary, Hyper-V on Windows Server 2025 will run on the X9QRi-F+ and can utilize its 4 CPU / large RAM architecture effectively for virtualization. The main limitations are those of the hardware generation: relatively lower per-core performance, no advanced NIC offloads, and missing modern security features. For many uses (running a farm of VMs for dev/test, hosting legacy enterprise applications, etc.) this setup will be more than sufficient. Just ensure you tweak the BIOS (enable VT-x/VT-d, ensure Hyper-Threading on or off as you prefer, power set to performance) and consider the networking/storage upgrades if needed to balance the system for virtual workloads.

Expected Performance on the X9QRi-F+ Platform

When evaluating overall performance, it's useful to compare the capabilities of the Windows Server 2025 + X9QRi-F+ combination against both modern systems and the baseline of this hardware. While we don't have a specific official benchmark of Windows Server 2025 on the X9QRi-F+, we can infer expected performance based on hardware specs and the improvements in the OS:

- **Compute Performance:** A fully loaded X9QRi-F+ with four Xeon E5-4667 v2 (for example) would have 48 cores @ 3.2 GHz. In terms of raw CPU, this was top-of-the-line in 2013. A single modern 2025-era CPU (like an Intel Xeon Platinum 8480+ or AMD EPYC) would significantly outperform it, especially in single-threaded tasks. However, for multi-threaded throughput, 48 Ivy Bridge cores still provide a lot of processing power. Windows Server 2025's efficient scheduling means that it will keep these cores busy. Real-world benchmarks: a similar 4x E5-4620 system running Windows Server 2012 scored ~45k in Geekbench multi-core ⁶⁹ ⁷⁰ (older Geekbench version), whereas a single new Xeon might score several times that. So one should set expectations: heavy workloads like modern encryption (where new CPUs have AES improvements and AVX-512) will run slower on this machine. But typical enterprise workloads (AD, file/print, moderate SQL databases, etc.) will still run at acceptable speeds as they are often I/O-bound or network-bound rather than pure CPU-bound.
- **Storage Throughput:** If sticking to SATA SSDs or HDDs, the max throughput per drive is ~500 MB/s for SATA SSD (or ~150 MB/s for a 7.2k HDD). The C602 has 2 SATA 6G ports that can each do ~550 MB/s with a fast SSD, and 4 SATA 3G ports that do ~280 MB/s max. In RAID or aggregate, you might see a few hundred MB/s. Windows Server 2025 doesn't change these physics, but it will make sure it can drive the devices at full speed. The NVMe improvement is noteworthy if an NVMe is used: e.g., a PCIe 3.0 x4 NVMe that typically does 3000 MB/s on older OS might do up to ~4800 MB/s on 2025

(60% gain) ¹⁶, limited by PCIe 3.0 x4 bandwidth (~3.94 GB/s). That 60% figure was likely measured with a very fast device and earlier OS inefficiencies; real-world improvements might be smaller for typical workloads, but it shows the OS can reduce overhead. For the X9, a realistic scenario: copying files between two SSDs or a network share could saturate the links – and if ReFS block clone kicks in (like copying a large VHDX on the same volume), it could complete almost instantly regardless of file size, which is a big win for admin operations.

- **Network Performance:** A single 1 Gbps link can transfer at ~115 MB/s. Windows Server 2025 will have no trouble hitting that ceiling. Where improvements might be seen is in latency and efficiency. For instance, under high packet-per-second loads, the networking stack in 2025 might use less CPU than 2012R2/2016 did. If your use case is a web server or similar, the lower jitter and improved TCP algorithms in 2025 could handle bursty traffic better. If iSCSI or SMB is used heavily, features like SMB Multichannel and SMB Direct (if a faster NIC with RDMA was added) would help. On this board as-is, a fair expectation is ~900-950 Mbps throughput on each NIC for large transfers, and the OS can handle multiple connections well. In a lab test, enabling SMB compression with LZ4 might raise CPU usage but could compress certain data enough to effectively exceed 1 Gbps (but that's data dependent).
- **Hyper-V VM Density:** If using the server for virtualization, an expected performance metric is how many VMs or what total load it can handle. Roughly, 48 cores can perhaps host on the order of 40-60 relatively light VMs (1 vCPU each) or a smaller number of heavier ones. The OS overhead for Hyper-V is minimal – Microsoft runs their Azure Stack HCI on similar multi-socket systems, and Windows is optimized to minimize hypervisor overhead. Windows Server 2025's support for huge VM sizes doesn't mean you can or should push that on this hardware, but it means any configuration you choose (like giving a VM 32 vCPUs and 256 GB RAM if needed) is supported by the OS, within the physical limits of 96 threads/1 TB.
- **Cooling and Power:** While not a performance metric, note that a fully loaded 4-CPU X9 system will consume a lot of power (potentially 600W+ under load) and generate significant heat. Windows Server 2025 has modern power management, but by default on a server, it tends to favor performance over power savings (especially if you select the "High Performance" power plan or the BIOS is set to OS Controlled but with performance bias). You can adjust the Windows power plan if you want to throttle the CPUs when idle (balanced plan etc.). However, for consistency, many server admins use "High Performance" mode to disable core parking and frequency scaling, ensuring steady performance. This will maximize throughput but also power usage. On older Xeons, the difference can be tens of watts, so it's a consideration: you can trade a bit of performance for efficiency if desired by using a balanced power plan.
- **Known Performance Issues:** As of release, Microsoft did acknowledge a few issues with Windows Server 2025 that could affect performance in edge cases. One relevant known issue: systems with **>256 logical processors** may experience BSODs or instability until patched ⁷¹. The X9QRi-F+ has at most 96 logical processors, so it stays well under that threshold – thus you shouldn't encounter that specific bug. Another issue was with **iSCSI boot** causing boot failures (a "boot device inaccessible" error in certain SAN boot scenarios) ⁷². This would only matter if you attempted to boot the OS from an iSCSI LUN. Most likely, you will install on local storage, so this is not a concern. These examples illustrate that while the OS is new, typical deployments on this hardware are not hitting the

problematic boundaries. Still, it's wise to keep Windows Update enabled and apply updates, as early patches will resolve such known bugs ⁷³ .

In summary, expected performance of Windows Server 2025 on the Supermicro X9QRi-F+ is solid for a system of this class. The OS will not be the limiting factor – the hardware will be. Think of it as giving a 2013 supercomputer a modern, efficient operating system: you get the benefits of new software optimizations, but you're still constrained by older CPU/memory speeds. The combination is appropriate for many tasks, especially if you're repurposing this board to extend its life running updated software. Just temper expectations for extremely demanding workloads (like high-end realtime analytics, heavy AI training, etc., which really need newer processors). For moderate enterprise workloads, lab environments, or virtualization of legacy apps, this setup should deliver reliable performance.

Known Issues and Limitations

When running Windows Server 2025 on the X9QRi-F+ platform, a few **known issues or limitations** should be noted:

- **Lack of Official Support:** As mentioned, Intel Xeon E5-v2 series CPUs are not officially supported in Server 2025. This means if you run into an OS bug, Microsoft support might not assist you unless you reproduce it on supported hardware. While the community and our testing indicate it works, you're essentially in an unsupported configuration. This is more of a *theoretical* limitation – day-to-day it likely won't matter, but it's worth acknowledging.
- **No TPM 2.0:** The X9QRi-F+ cannot accommodate a TPM 2.0 module ⁶⁵ . This limits certain security scenarios. For instance, if you wanted to implement **BitLocker Drive Encryption** with a TPM protector, you can't – you'd have to use a password or USB key for BitLocker instead. Also, features like **Device Health Attestation** or Credential Guard's highest mode require TPM 2.0. The practical impact is that the server might not meet some compliance standards that require TPM-based attestation. If this is a lab or non-critical deployment, it may not matter. If it's going into production in an environment with strict security requirements, this could be a show-stopper for using this older board.
- **Older BIOS and UEFI Limitations:** The motherboard's UEFI, while functional, is not as advanced as modern ones. Beyond the NVMe boot issue discussed, it may also lack some quality-of-life features. For example, many X9 boards do not have a graphical UEFI interface (just text), and they might not fully support newer peripheral boot options. There have been instances where certain newer GPUs in X9 systems won't display output until OS loads (due to GOP driver issues in old UEFI). If you plan to install a modern GPU for compute, be mindful that you might need to update the GPU's firmware to one that supports legacy mode or simply operate headless. Additionally, the board might not support "UEFI HTTP Boot" or other modern network boot standards that Windows Server 2025 can do – only legacy PXE. These are minor limitations, but if you intended to, say, use Windows Deployment Services with UEFI HTTP boot, that won't work on this hardware.
- **Power Consumption and Heat:** While not an OS issue, running this board fully loaded with Windows Server 2025 means you'll be drawing significant power even at idle compared to newer hardware. The OS itself might try to idle cores, but the power gating on older Xeons is not as effective as on newer ones. This means the system could run hotter and noisier. Ensure the server's

cooling (fans, heatsinks) and power supply are in good shape. Thermal management is partly handled by the BMC/BIOS (fan speed ramp-up based on sensor thresholds). Windows can signal certain ACPI states but generally Supermicro's cooling is firmware-driven. Just watch that under heavy multi-core loads the CPUs don't throttle – if they do, consider increasing cooling or checking that thermal paste etc., is appropriate.

- **Driver Support Ceases:** While all necessary drivers are available now, keep in mind that future updates (beyond 2025) won't be targeting this hardware. Intel and others have long since stopped updating drivers for the X9's components. If a future Windows 2025 cumulative update were to, hypothetically, introduce an incompatibility with the old Intel AHCI driver or similar, there's a slim chance you could hit an issue. This is unlikely, but it's part of running an OS on decade-old gear. The good news is the drivers in Windows are stable – e.g., the in-box e1000e driver for the i350 NIC is mature, and the AHCI driver is Microsoft's own which is well-tested.
- **Blue Screen on >256 Threads:** Not applicable to this board, but worth noting as an OS bug: if you had a server with more than 256 logical processors, Server 2025 could crash (this is a known issue to be fixed) ⁷⁴. With 96 LP, the X9QRi-F+ is safe from this, so it's actually an example where this older hardware avoids a new software bug that only huge core count systems face!
- **iSCSI Boot Bug:** Again, only relevant if you try something exotic like network booting the OS from an iSCSI target. Windows Server 2025 has a known issue where that might fail with a BSOD or inaccessible boot device error ⁷². This board could theoretically PXE boot WinPE and then iSCSI mount a LUN to boot Windows – but that's not a common scenario. Most will install on local disk. So this is a known OS issue rather than a board issue, but we include it for completeness.
- **Maximizing Upgrades:** The X9QRi-F+ is already at its CPU and RAM limits with v2 CPUs and DDR3. Unlike some slightly newer boards, you can't upgrade beyond that (no v3 CPU support, etc.). So the limitation is that you're at the ceiling of this board's performance. If Windows Server 2025 sparks you to do more (containers, more VMs, etc.) you might quickly find the system saturated where a modern platform would not be. This is just a consideration for planning: if this is for testing or interim use, fine – but for long-term production use, the lack of future hardware upgrade path (short of replacing the whole server) is a limitation to be aware of.

Notwithstanding these limitations, there aren't any show-stopping *incompatibilities* reported for Windows Server 2025 on this board. Users have successfully run even Windows 11-based OS on similar X9 systems ⁶¹. The experience is generally stable. Any issues that do arise (random freezes, etc.) are likely to be hardware-related (aging components, need for BIOS update, etc.) rather than OS bugs given the relatively standard hardware in the X9 (Intel reference platform for the most part).

Recommendations for Optimal Stability and Performance

To ensure the best experience running Windows Server 2025 on the Supermicro X9QRi-F+, consider the following best practices, BIOS tweaks, and firmware updates:

- **Update System Firmware:** As stated, update to the latest BIOS release for the X9QRi-F+ (check Supermicro's website). Also update the BMC/IPMI firmware to the newest version. Newer BIOS

revisions often improve CPU microcode, memory compatibility, and fix errata that could cause system hangs under stress. Administrators have pointed out that unexplained stability issues on Supermicro boards can often be resolved by bringing firmware up-to-date ⁶³. Since this board is old, the “latest” BIOS is from 2015, but make sure you have that rather than an initial 2013 version.

- **Enable Relevant BIOS Settings:** Enter the BIOS setup and verify:
 - **Virtualization Settings:** VT-x (Intel Virtualization Tech) = Enabled; VT-d (Directed I/O) = Enabled. This ensures Hyper-V and device passthrough function properly.
 - **Memory Settings:** If there’s an option for “Node Interleaving,” **Disable** it (this keeps NUMA nodes separate, which is better for performance). If Node Interleaving is enabled, it would interleave memory across NUMA nodes, potentially hurting performance for local memory access. By default it’s usually disabled on multi-socket Supermicros, but good to confirm.
 - **Power/Performance:** Set the BIOS power management to “**Performance**” or **OS Controlled (and then in Windows set High Performance plan)**. On some X9 boards, there is a “Power Efficient” vs “Performance” setting. The latter will minimize CPU C-state stepping and give more consistent performance (at the cost of power). If you prefer the CPUs throttle when idle, you can choose Balanced, but for a server workload performance mode is often recommended.
 - **Hyper-Threading:** Ensure Hyper-Threading is **Enabled** (unless you have a specific reason to disable it, such as certain licensing or security considerations). Hyper-Threading will give you up to ~20-30% throughput boost for many workloads. There were past concerns about side-channel attacks on HT (like L1TF vulnerability), but those can be mitigated via patches rather than disabling HT in most cases.
 - **Boot Mode & Secure Boot:** Set Boot Mode = UEFI. Enable Secure Boot and load the default keys (as long as you intend to run Windows or other signed OS). This will help protect the boot process ⁶⁴. If you run into any boot issues after enabling Secure Boot (for example, if using some unsigned kernel drivers or utilities in pre-boot), you might need to add keys or disable it, but generally Windows Server works with Secure Boot on.
 - **Above 4G Decoding:** If you plan to install devices that use large BAR memory ranges (like a GPU or certain NVMe RAID cards), check if the BIOS has an option for “Above 4G Decoding” or “64-bit PCI decoding” and **enable** it. This allows the BIOS/OS to map MMIO space above 4GB address range, which is needed if you have devices that collectively require more than 4GB of address space (common if a large GPU with 16GB VRAM is installed). This setting is often found in later-gen boards; some X9 BIOS have it hidden or not available. If it’s present, enabling it preemptively can avoid issues with devices not showing up in Windows due to address space collisions.
- **Install Latest Drivers (if available):** After OS install, check Device Manager. If any devices are unnamed (e.g., “Base System Device”), install the Supermicro or Intel chipset INF to properly tag them. For NICs, consider installing the Intel PROSet drivers for Windows Server (even if 2025 isn’t listed, the latest available for 2019/2022 should work). This can sometimes provide adapter teaming software or advanced diagnostics. For storage, if using RAID, install the appropriate Intel/LSI drivers and management utility (like Intel RST CLI or MegaRAID Storage Manager) to monitor array health. While Microsoft’s in-box drivers are stable, vendor drivers can offer performance tuning or full feature support.

- **Verify Event Logs and Device Manager:** Keep an eye on the Windows Event Viewer for any warnings or errors that might indicate a driver or hardware issue (for example, storport timeouts, corrected ECC errors, etc.). Also verify that all 4 CPUs and full memory are recognized by the OS (you can check System Information or Task Manager > Performance). If something looks off (like only 3 CPUs showing), that could indicate a CPU or socket not initialized – re-seat CPUs or adjust BIOS settings accordingly.
- **Stress Test the System:** After installation and setup, it's wise to run some stress tests to ensure stability under load. For example, run Hyper-V with some test VMs doing CPU+memory stress, or use tools like Prime95, memtest, or disk benchmarking (CrystalDiskMark on your drives, iperf for network). This can reveal any cooling insufficiencies or configuration issues (like a mis-seated RAM module causing correctable ECC errors under load, etc.). It's better to identify and address such issues early.
- **Plan for Redundancy:** This is more operational advice – if this server will host important workloads, remember that the hardware is old and components (PSUs, drives, fans) can fail. Ensure you have proper cooling (all fans operational – this board has 10 fan headers and expects a robust chassis cooling ⁷⁵ ⁷⁶). Consider keeping a cold spare or at least have a backup strategy for the VMs/data. Windows Server 2025 can use Storage Replica to copy data to another server or Azure for backups. Utilize those OS features to mitigate the risk of hardware failure.
- **Leverage New OS Features Wisely:** Make use of the OS enhancements to get the most out of the hardware. For example, if running VMs, use **VM Start Order Priority and Automatic Start Delays** to stagger VM boot if you have many (to avoid power spike and disk thrash on boot). Use **Storage QoS** if multiple VMs share a disk system, to prevent one VM from saturating the I/O. If this server is an all-in-one, use **Nested Virtualization** if you want to run Hyper-V inside a VM (yes, Server 2025 supports nesting even on older Intel CPUs that have VT-x/EPT – useful for lab scenarios). Also, enable **Windows Admin Center** for a modern management experience – you can install it on Windows Server 2025 (it's a separate download or use via Azure Arc as mentioned). WAC gives a nice GUI to monitor performance, manage updates, etc., which is helpful if you don't want to RDP into the server regularly.
- **Security Updates:** Continue to apply Windows Updates. Even though the hardware is old, the OS updates (especially security patches) are crucial. Windows Server 2025 will be supported with updates into 2029 (mainstream) and 2034 (extended) ⁷⁷ . Staying patched will keep the server secure and stable. If any update ever seems to cause an issue (rare, but e.g. some early 2025 update reportedly caused DC connectivity issues ⁷⁸), you can always roll it back, but generally, trust the update process.

By following the above recommendations, you can achieve a stable and well-performing system. Many IT professionals have successfully modernized older servers like the X9QRi-F+ by running the latest software on them – the key is to mind the details (firmware, configuration, realistic expectations). In conclusion, Windows Server 2025 can breathe new life into the Supermicro X9QRi-F+ platform, providing access to current-generation features and security on this veteran hardware, as long as one carefully addresses the compatibility and optimization points discussed.

Table: Windows Server 2025 vs. Supermicro X9QRi-F+ – Key Compatibility Highlights

Aspect	Windows Server 2025 Capability	Supermicro X9QRi-F+ Hardware	Compatibility Notes
CPU Support	Up to 2048 logical processors per host ² (64+ sockets) Supports latest Intel Xeon Scalable & AMD EPYC families ⁶⁰	4× Intel Xeon E5-4600 v2 (12 cores each, 48 cores / 96 threads max) ⁵⁷ (Ivy Bridge, 22nm, launched 2013)	<i>Not officially on support list</i> , but meets 64-bit, VT-x/SLAT requirements. No modern AVX2/AVX-512, which limits some performance. OS can use all 96 threads.
Memory	Supports up to 4 PB RAM per host ² Enhanced memory management for large and NUMA systems	32 DIMM slots, up to 1 TB DDR3 ECC (1600/1866 MT/s) ⁴⁸ Quad-channel per CPU (NUMA nodes x4)	OS can address full 1 TB with ease. NUMA optimizations in OS will maximize local memory use. Older DDR3 bandwidth is a limitation for memory-intensive apps.
Storage & I/O	NVMe-optimized (up to +60% IOPS vs 2022) ¹⁶ ReFS with block cloning, dedup, compression ²⁰ ²² Storage Spaces Direct (S2D) with thin provisioning ²³	6 onboard SATA ports (2 @6Gb/s + 4 @3Gb/s) ¹⁸ PCIe 3.0 slots for add-on NVMe or SAS controllers No native M.2/U.2 support (NVMe via adapter only)	SATA SSD/HDD performance is supported (AHCI/RAID). OS benefits like ReFS fast clone help with VHDX on any storage. NVMe improvements require NVMe drives on PCIe (BIOS can't boot NVMe without workaround). S2D features work if using add-on HBAs and multiple disks, but 1GbE networking is a bottleneck for cluster use.
Networking	Improved TCP/IP stack, low-latency optimizations ⁵¹ Supports 1GbE to 100GbE NICs, SR-IOV, RDMA SMB over QUIC and SMB compression for efficient file transfers ²⁷	2× 1 GbE LAN (Intel i350-AM2) ⁴⁷ 1× IPMI 2.0 LAN (Realtek PHY for BMC) PCIe slots for higher-speed NICs (10G/25G etc.)	Onboard 1GbE works with in-box drivers (full features except SR-IOV). Throughput max ~1 Gbps per port; OS easily handles that. SR-IOV and RDMA not available on i350 – require add-in NIC. SMB over QUIC can be used on 1GbE, but primarily benefits remote access scenarios.

Aspect	Windows Server 2025 Capability	Supermicro X9QRi-F+ Hardware	Compatibility Notes
Security	Requires UEFI & TPM2.0 for “secured-core” features Credential Guard on by default (if supported) ²⁸ VBS enclaves, Hypervisor-protected code integrity ¹² Secure Boot support, SMB security hardening ⁶⁴ ²⁷	UEFI 2.3.1 BIOS with Secure Boot, ACPI 4.0 ³⁰ Optional TPM 1.2 module (no TPM 2.0 available) ⁶⁵ BMC provides out-of-band management (no direct OS security impact)	Secure Boot can be enabled on this board (protect boot process) ⁶⁴ . Lacks TPM 2.0 – cannot utilize TPM-dependent features (BitLocker with TPM, Device Health Attestation, etc.). Credential Guard/VBS may not activate due to no TPM & older CPU lacking MBEC – OS will fall back gracefully or disable those features. Core OS security (firewall, Windows Defender, etc.) fully functional.
Virtualization	Hyper-V with 2048 vCPUs per VM, 240 TB/vm ³ GPU Partitioning and live migration of GPU-backed VMs ⁹ Hyper-V Failover Clustering (incl. Workgroup mode) ¹³ Accelerated Networking (SR-IOV) for VMs ⁷	Supports Hyper-V role (VT-x/EPT present) Up to 96 logical processors for host, exposed to VMs No onboard GPU for GPU-P (requires add-in GPU) NICs do not support SR-IOV (Intel i350 limitation)	Hyper-V runs well; can create and run many VMs within CPU/RAM limits. No SR-IOV means higher vSwitch overhead (moderate CPU cost for VM networking). Live Migration supported over 1GbE (recommend enabling compression or multi-channel for faster VM moves). GPU acceleration in VMs requires GPU hardware; otherwise VMs use CPU-only rendering.
Remote Management	Windows Admin Center (WAC) integration – manage via web GUI ³⁷ Azure Arc enabled – cloud-based management and monitoring ⁷⁹ ³⁷ PowerShell 7 and OpenSSH for remote CLI admin ⁴³	Dedicated IPMI module with KVM, power control ⁴⁷ Out-of-band management independent of OS Standard IPMI 2.0 commands and web interface	OS-level remote management (WAC, PS Remoting) works provided network access. IPMI allows remote access even if OS is down – use for hardware management (fan speeds, BIOS settings, reinstall OS etc.). Combine both: e.g., use IPMI for lights-out management, and WAC/Arc for day-to-day OS management. No conflict between them.

Sources: Official Microsoft documentation and release notes for Windows Server 2025 features ² ¹⁶ ⁸⁰ , Supermicro X9QRi-F+ specifications ⁵⁷ ⁴⁷ , and user reports on compatibility (Reddit) ⁶⁵ ⁶¹ .

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